

Engr. Anas Osman

PhD Researcher in Geodesy & Seismology | Surveying Engineer

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Introduction

I am a PhD researcher in geodesy and seismology using InSAR and GNSS to study how the Earth's crust moves before, during and after earthquakes, how faults slip and how earthquakes influence nearby faults, with additional experience in remote sensing, GIS and gravimetric geoid modelling from my surveying and geomatics engineering studies.

Personal Details

Website	osmananas.github.io
Address	No. 340, Xudong Avenue, Wuhan City, Hubei Province, China.
Nationality	Sudanese.
Date of birth	3 February 1995.
Marital status	Single.
Place of birth	Khartoum, Sudan.
Personal interests	Learning languages, Travel and cultural exchange; reading about science and technology; photography; football; Hiking and field exploration.

Education

Sep 2023 – Jul 2027 (expected)	PhD, Geodesy and Surveying Engineering University of Chinese Academy of Sciences (UCAS), Beijing, China. Doctoral research host since Jan 2024: Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences (CAS), Wuhan, China. <i>PhD qualification examination passed in June 2024.</i>
Jul 2020 – Aug 2022	Master of Engineering, Geomatics Engineering Institut Teknologi Sepuluh Nopember (ITS), Surabaya, Indonesia. <i>Cum laude; CGPA: 3.86/4.00. Department of Geomatics Engineering.</i>
2012 – 2017	BSc (Honours), Surveying Engineering Omdurman Islamic University, Omdurman, Sudan. <i>Second Class, Division One. Faculty of Engineering Sciences.</i>

Research Interests

- **Earthquake-cycle geodesy and fault interactions:** interseismic strain, coseismic slip and postseismic deformation using InSAR and GNSS, including afterslip, aftershock triggering and Coulomb stress transfer.
- **Remote sensing and GIS:** satellite imagery, SAR interferometry, deformation time series, land subsidence and geospatial analysis.
- **Physical and satellite geodesy:** gravimetric geoid modelling, gravity-field analysis and evaluation of global geopotential models.

Research Experience

Jan 2024 – Present	Doctoral Research, Geodesy and Seismology Innovation Academy for Precision Measurement Science and Technology, CAS. • Investigate earthquake-cycle deformation using InSAR and GNSS observations, with published work on the Wushi and Kamchatka earthquake sequences. • Model coseismic slip and afterslip, and evaluate earthquake interactions using Coulomb stress changes and aftershock distributions. • Four additional research manuscripts are in progress or under review.
Jul 2020 – Aug 2022	Master's Research, Gravimetric Geoid Modelling Institut Teknologi Sepuluh Nopember (ITS). • Developed a regional gravimetric geoid model for Sudan using least-squares collocation. • Evaluated global geopotential models and digital elevation models against terrestrial gravity and GNSS/levelling observations.

Research Publications

Five journal articles and three conference papers. **Anas Osman** is highlighted in each author list.

Journal Articles

- 2026 **Interplay of Coseismic Rupture, Afterslip, and Aftershocks Associated with the 2024 Mw 7.1 Wushi Earthquake**
Osman, A., Guo, R., Tang, X., Zhang, Y., Zhang, B., Sun, H., & Abdelaal, M. I. (2026). *Remote Sensing*, 18(16), 2739.
doi:10.3390/rs18162739
- 2026 **Interseismic, Coseismic, and Early Postseismic Slip Associated With the 2025 Mw 8.8 Kamchatka Earthquake**
Xiao, Y., Wang, J., Zhang, Y., Zhang, W., Zhou, J., Sun, H., Zhao, Y., **Osman, A.**, & Guo, R. (2026). *Geophysical Research Letters*.
doi:10.1029/2025GL121473
- 2026 **Multi-Temporal Identification and Analysis of Land Surface Deformation with Land Use Land Cover in Makassar City**
Hamzah, S., Wijayanti, R. F., Anjasmara, I. M., Sabri, L. M., & **Osman, A.** (2026). *Indonesian Journal of Urban and Environmental Technology*, 9(1), 40–60.
doi:10.25105/urbanenvirotech.v9i1.23368
- 2025 **Regional evaluation of global geopotential models and three types of digital elevation models with ground-based gravity and GNSS/levelling data using several techniques over Sudan**
Osman, A., Elsaka, B., & Anjasmara, I. M. (2025). *Journal of Applied Geodesy*, 19(1), 25–48.
doi:10.1515/jag-2024-0006
- 2021 **Accuracy Investigation of the Three-Point Resection Method Through the Distribution of Control Points Across Four Quadrants (Based on BSc thesis)**
Osman, A., Mabrouk, A. M. A., Mahjoub, A. M. A., Elkhailifa, E. A. M., & Elhag, A. A. (2021). *Journal of Marine-Earth Science and Technology*.
doi:10.12962/j27745449.v2i2.101

Conference Papers

- 2023 **Determination of a new gravimetric geoid modelling for Sudan using the least-squares collocation technique**
Osman, A. S. M., & Anjasmara, I. M. (2023). *IOP Conference Series: Earth and Environmental Science*, 1127(1), 012014.
doi:10.1088/1755-1315/1127/1/012014
- 2021 **Assessment of high-degree reference models and Recent Goce/Grace Global Geopotential Models over Sudan based on the GPS/Leveling data**
Osman, A. S. M., Anjasmara, I. M., Ruby, A., & Udama, Z. A. (2021). *IOP Conference Series: Earth and Environmental Science*, 936(1), 012035.
doi:10.1088/1755-1315/936/1/012035
- 2021 **Geoid Modelling of Kalimantan Island using Airborne Gravity Data and Global Geoid Model (EGM2008)**
Udama, Z. A., Anjasmara, I. M., Pahlevi, A. M., & **Osman, A. S. M.** (2021). *IOP Conference Series: Earth and Environmental Science*, 936(1), 012029.
doi:10.1088/1755-1315/936/1/012029

Professional Experience

- Sep 2022 – **Adjunct Lecturer (Part-time)**
Nov 2024 Department of Surveying Engineering, Omdurman Islamic University, Sudan.
• Lectured undergraduate (BSc) courses and supervised final-year projects and undergraduate theses.
- Dec 2017 – **Surveying Engineer (Contract)**
Dec 2018 Dar Consult, Khartoum, Sudan.
• Worked as a consultant surveying engineer on road and bridge construction projects.
• Modified engineering designs and supervised contractors during implementation at construction sites.

Technical Skills

Geodesy and seismogeodesy	InSAR processing and time-series analysis (SBAS, PS-InSAR); GNSS data processing and deformation analysis; deformation monitoring; coseismic and postseismic modelling; finite-fault slip inversion; Coulomb stress modelling; elastic dislocation modelling; gravimetric geoid modelling; least-squares adjustment and uncertainty analysis.
Remote sensing and GIS	Satellite image processing; SAR interferometry; LiDAR point-cloud processing; digital elevation model (DEM) generation; geospatial data processing and analysis; ArcGIS, QGIS and SNAP.
Programming and scripting	MATLAB, Python (NumPy, SciPy, Pandas, Matplotlib) and Bash shell scripting; Git for version control; Linux/Unix environments.
Machine learning and AI	Machine learning and artificial intelligence applied to geoscience data; scikit-learn, TensorFlow and PyTorch.
Research software	GMTSAR, ISCE, MintPy, StaMPS, PSGRN/PSCMP, SDM, SlipSolve, GRAVSOFT, Coulomb, GAMIT/GLOBK, GipsyX.
Visualisation and documentation	GMT and PyGMT; Inkscape; Matplotlib; Surfer; Google Earth Pro; LaTeX and Overleaf; Microsoft Office.
surveying Engineering	AutoCAD Civil 3D; GNSS RTK; total station; levelling; topographic surveys; setting out; network adjustment.

Soft Skills

Time management, problem-solving, Strong work ethic, Presentation skills, teamwork, Quick learning and adaptability, Research and analytical skills, Creativity, Communication skills.

Languages

Arabic	Native or bilingual proficiency.
Nubian	Native or bilingual proficiency.
English	Full professional proficiency.
Chinese	Elementary proficiency.
Indonesian	Elementary proficiency.

Academic Service and Professional Registration

Peer-review Engagement

Reviewed manuscripts for several academic journals.

May 2023 –	Reviewer
Present	Journal of Marine-Earth Science and Technology.
Apr 2018	Graduate Engineer (Surveying), Sudan Engineering Council
	Registration number: EC/ER/GE/55449.

Honours, Awards and Scholarships

Aug 2025	Excellent Poster Award Joint International Symposium on Geodesy and Geodynamics and the 13th International Workshop on Multi-observations and Interpretations of Xizang, Xinjiang and Siberia; Xining, China.
Jul 2023	ANSO-CAS-TWAS/UNESCO PhD Scholarship ANSO Scholarship for Young Talents; ANSO, CAS and TWAS/UNESCO.
Aug 2022	Master of Engineering with Cum Laude Institut Teknologi Sepuluh Nopember; Geomatics Engineering, CGPA 3.86/4.00.
Jun 2019	KNB Scholarship for Master's Studies Indonesian Government; master's studies in Geomatics Engineering at ITS.
Jul 2017	Academic Excellence Award, Surveying Engineering Omdurman Islamic University: highest academic performance in semester 10 and second-best overall result in the fifth academic year (2016/2017).

Academic Conferences and Presentations

- Aug 2026 **14th International Workshop on TibXS**
Multi-observations and Interpretations of Xizang, Xinjiang and Siberia; Kunming, China, 10–14 August 2026. Presenter.
- Aug 2025 **Joint Geodesy and Geodynamics Symposium and 13th TibXS Workshop**
Xining, China, 1–4 August 2025. Poster presentation; recipient of the Excellent Poster Award. Presenter
- Oct 2024 **GGOS Days 2024**
Global Geodetic Observing System, International Association of Geodesy. Participant.
- Jul 2022 **Geomatics International Conference (GEOICON 2022)**
Institut Teknologi Sepuluh Nopember; Geospatial Technology for Mapping the Future (3D Cadastre Development). Conference contribution on Sudan gravimetric geoid modelling using least-squares collocation. Presenter.
- Jul 2021 **Geomatics International Conference (GEOICON 2021)**
Institut Teknologi Sepuluh Nopember; Geospatial Technology for Mapping the Future (Solutions for Hazards and Disaster Management). Presenter.
- Oct 2019 **Joint International Conference on Civil, Environmental and Geo Engineering**
Department of Geophysical Engineering, Institut Teknologi Sepuluh Nopember. Presenter.

Selected Training and Professional Development

Satellite Observations, Remote Sensing and GIS

- Nov 2025 **COMET InSAR Training Workshop**
Centre for Observation and Modelling of Earthquakes, Volcanoes and Tectonics, UK.
- Apr 2024 **Advanced InSAR and Data Analytics for Civil Engineering**
AGS (Hong Kong); construction-site risk reduction and long-term asset maintenance.
- Mar/Jun 2024 **From Space to Screen: Satellite Imagery in ArcGIS**
Esri; accessing and tasking satellite imagery.
- Mar 2024 **NOAA SatBathy: Satellite-Derived Bathymetry**
NOAA; automating bathymetry from satellite data.
- May 2019 **Cartography**
Esri.

GNSS, Gravity and Seismology

- 2020 – 2024 **GNSS Positioning, Time Series and Observation Planning**
NOAA National Geodetic Survey / COMET: GNSS noise evaluation, orbit determination, multi-GNSS positioning, space-weather impacts, antenna calibration, survey planning and Southern Louisiana GNSS network results.
- 2024 **EarthScope Seismology Skill Building Workshop**
EarthScope Consortium / Miami University; participated in Linux, shell scripting, GMT and introductory SAC tutorials.
- 2022 – 2024 **Geoid and Gravity Training**
NOAA National Geodetic Survey / COMET: Geoid Monitoring Service (GeMS), Gravity for Geodesy II: Applications and Progress of GEOID2022.

Additional NOAA/NGS webinars

OPUS; CORS installation, maintenance, network modernization and map products; National Spatial Reference System modernization; absolute gravity; GEOID18 improvements and geoid change in Alaska; GPS observations of wetland resilience to sea-level rise.

Research Practice and Engineering

- 2024 – 2025 **Peer Review, Open Science and Scientific Writing**
Researcher Academy, University of Nottingham: peer review, manuscript preparation, publication ethics, research metrics and journal selection. Also attended Systematic Review for Beginners, Systematic Reviews Network (Jun 2024).
- Jan/Feb 2019 **Engineering Software and Quantity Surveying**
Zoohal Engineering and Training Center: AutoCAD Civil 3D, Primavera P6, PlanSwift, earthworks, quantity surveying in Excel, project costs, contracts and specifications.
- Aug–Sep 2014 **Practical Surveying Training**
Sudan Survey Authority; training in digital leveling, Leica total station, GPS fundamentals, data transfer, and field surveying.